<table>
<tr><th>No.</th><th>Solution</th><th>Remarks</th></tr>
<tr><td>1(a)</td><td>As the cyclist pedals forward, the ground exerts a forward force on the bicycle that is equal in magnitude and opposite to the frictional forces opposing the motion of the cyclist. Thus there is no resultant force acting on the cyclist, and the cyclist will continue in his uniform state of motion at constant speed in a straight line according to Newton's first law of motion.</td><td>[1] for correct explanation

[1] for no resultant force</td></tr>
<tr><td>1(b)</td><td>Making v the subject,
$v=\sqrt{\dfrac{2F}{c_D\rho A}}$
$=\sqrt{\dfrac{2(23)}{(0.88)(1.2)(0.32)}}$
$=11.667$
$\approx 11.7\text{ m s}^{-1}$
$\dfrac{\Delta v}{v}=0.5\left(\dfrac{\Delta F}{F}+\dfrac{\Delta c_D}{c_D}+\dfrac{\Delta\rho}{\rho}+\dfrac{\Delta A}{A}\right)$
$=0.5\left(\dfrac{2}{23}+\dfrac{0.01}{0.88}+\dfrac{0.1}{1.2}+\dfrac{0.02}{0.32}\right)$
$=0.122$
$\Delta v=(0.122)(11.667)$
$=1.42$
$\approx 1\text{ m s}^{-1}$ (1 s.f.)

Hence, $v=(12\pm 1)\text{ m s}^{-1}$</td><td>[1] for correct v

[1] for correct expression of $\dfrac{\Delta v}{v}$ and numerical substitution

[1] for correct Δv in 1 s.f.

[1] for correct $v\pm\Delta v$ to an appropriate no. of s.f.</td></tr>
<tr><td>1(c)(i)</td><td>Let the driving force be F_D and the displacement be s.

Work done, $W=F_D\times s$

Since net force = 0, F_D = drag force = 23 N

power output
$=$ rate of work done
$=\dfrac{dW}{dt}$
$=\dfrac{d(23s)}{dt}$
$=23\dfrac{ds}{dt}$
$=23v$</td><td>[1] for expression and substitution

[1] for correct answer</td></tr>
</table>

No.	Solution	Remarks
1(c)(ii)	Power, $P = Fv$ $= 23 \times 11.667$ $= 268\ \text{W}$	[1] for correct numerical substitution and answer